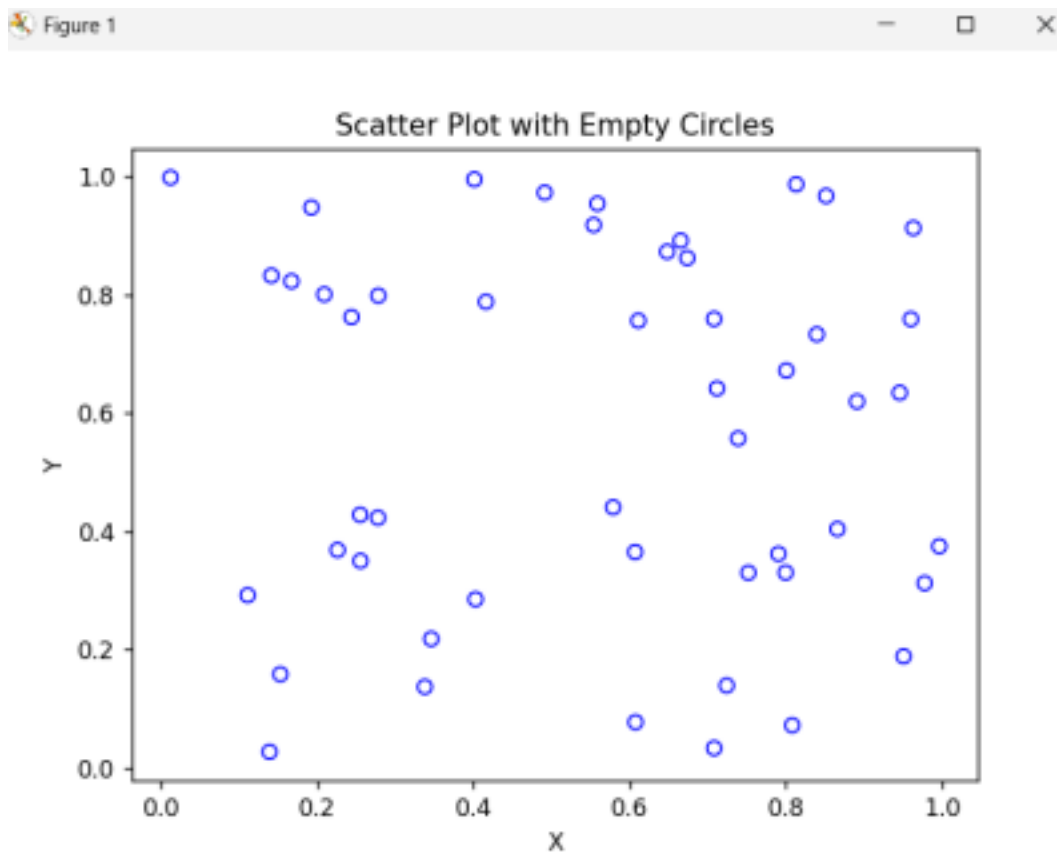


33. Write a Python program to draw a scatter plot with empty circles taking a random distribution in X and Y and plotted against each other. Program:

```
import matplotlib.pyplot as plt
import numpy as np
x = np.random.rand(50)
y = np.random.rand(50)
plt.scatter(x, y, facecolors='none', edgecolors='blue')
plt.xlabel("X")
plt.ylabel("Y")
plt.title("Scatter Plot with Empty Circles")
plt.show()
```



Output: 

34. Write a Python program to draw a scatter plot using random distributions to generate balls of different sizes.

Program:

```
import matplotlib.pyplot as plt
```

```
import numpy as np
```

```
x = np.random.rand(50)
```

```
y = np.random.rand(50)
```

```
sizes = np.random.randint(20, 300, 50)
```

```
plt.scatter(x, y, s=sizes)
```

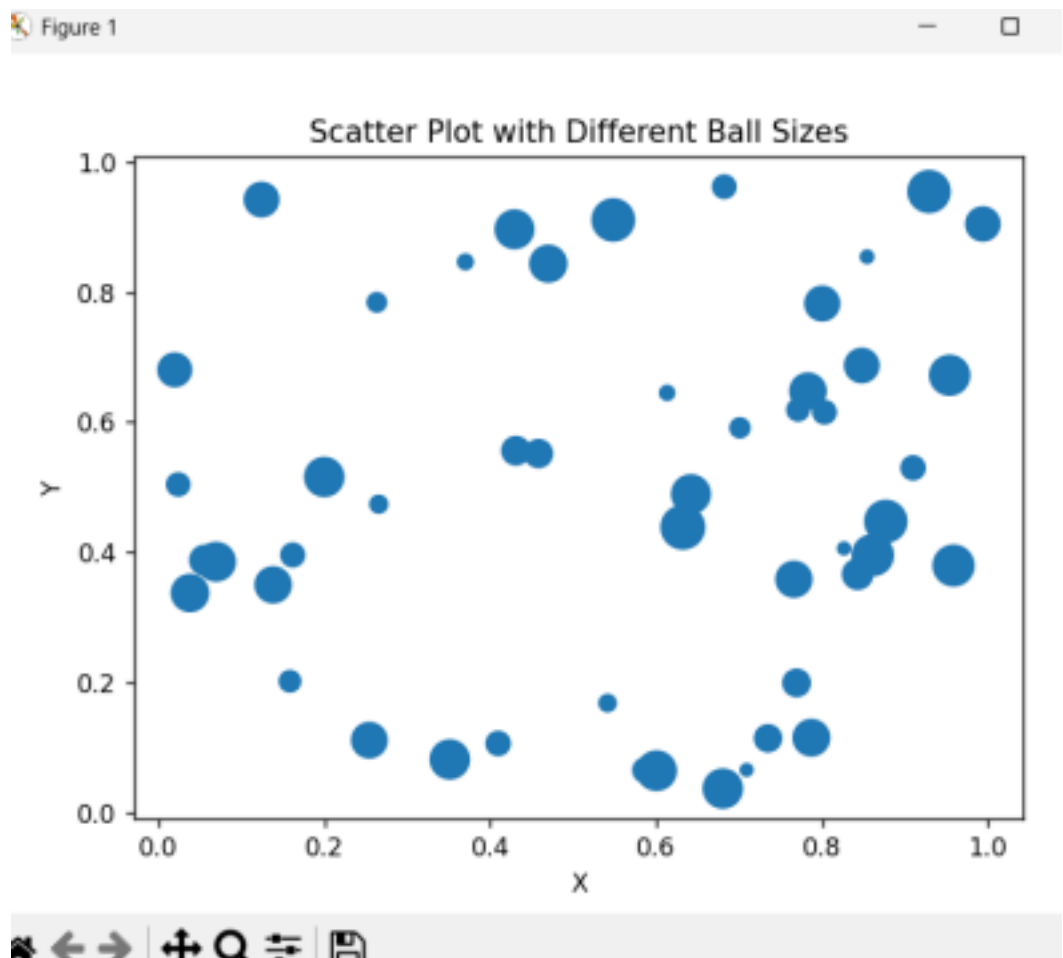
```
plt.xlabel("X")
```

```
plt.ylabel("Y")
```

```
plt.title("Scatter Plot with Different Ball Sizes")
```

```
plt.show()
```

Output:



35. Write a Python program to draw a scatter plot comparing two subject marks of Mathematics and Science. Use marks of 10 students. Program:

```
import matplotlib.pyplot as plt
```

```
math_marks = [88, 92, 80, 89, 100, 80, 60, 100, 80, 34]
```

```
science_marks = [35, 79, 79, 48, 100, 88, 32, 45, 20, 30]
```

```
marks_range = [10, 20, 30, 40, 50, 60, 70, 80, 90, 100]
```

```
plt.scatter(marks_range, math_marks, label="Mathematics") plt.scatter(marks_range,  
science_marks, label="Science")
```

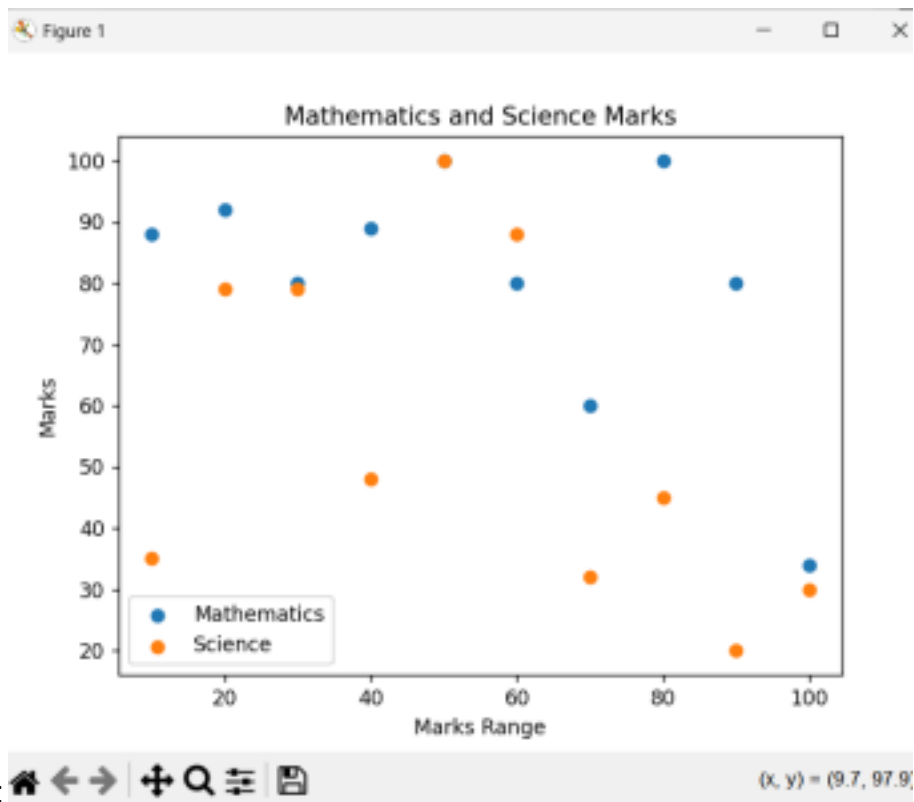
```
plt.xlabel("Marks Range")
```

```
plt.ylabel("Marks")
```

```
plt.title("Mathematics and Science Marks")
```

```
plt.legend()
```

```
plt.show()
```



Output: (x, y) = (9.7, 97.9)